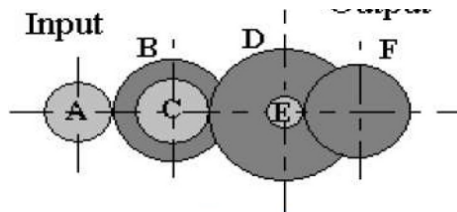


Tutorial 7: Gears

In-class activities

1. A pair of gears consist of a 20 teeth pinion meshing with a 120 teeth gear. The module is 4 mm. Calculate the pitch circle of the pinion and gear, the center distance, and the gear ratio.
2. Two modules of 3 mm gears are to be mounted on a center distance of 360 mm. The speed ratio is $1/3$. Determine the number of teeth in each gear.
3. Calculate the speed ratio for the compound chain shown below. If the input gear rotates clockwise, in which direction does the output rotate?

Gear	A	B	C	D	E	F
No. of Teeth	20	100	40	100	10	100



Theory

1. Describe the different types of gears and their applications.
2. Discuss the importance of gear material selection and lubrication.
3. Explain possible failure modes in gears and how to prevent them.
4. Explain the concept of gear ratio and how it affects torque and speed.
5. Define gear train and provide examples of its use in mechanical systems.

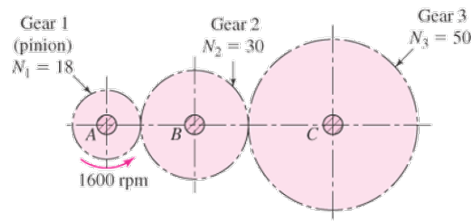
Calculations

Question 1

The gears shown in Figure below have a module of 6 mm and a 20° pressure angle. Determine, and show on a free-body diagram,

- The tangential and radial loads on each gear.
- The reactions on shaft C.

Given: Driver gear 1 transmits 80 kW at 1600 rpm through the idler to gear 3 mounted on shaft C.



Solution:

Given: $P = 80 \text{ kW}$

$N_1 = 1600 \text{ rpm}$

module, $m = 6 \text{ mm}$

pressure angle, $\phi = 20^\circ$

From the figure, we have

$N_1 = 20$, $N_2 = 30$, $N_3 = 50$

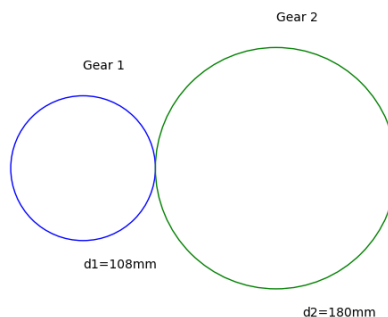
Calculating the pitch diameter of each gear

$$d_1 = m \times N_1 = 108 \text{ mm}$$

$$d_2 = m \times N_2 = 180 \text{ mm}$$

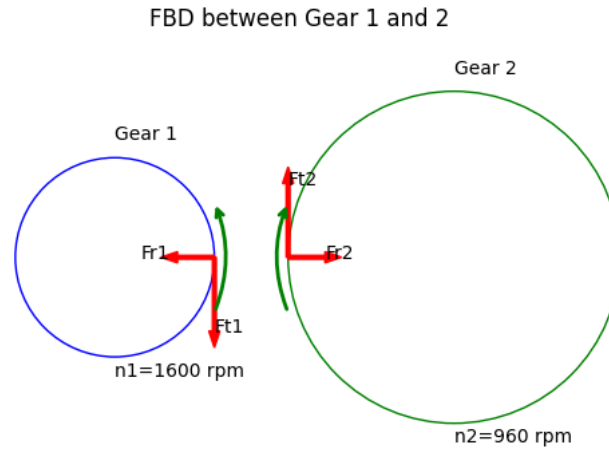
$$d_3 = m \times N_3 = 300 \text{ mm}$$

Gear 1 and 2



Draw FBD between gear 1 and gear 2

1. Identify which one is driver and driven gear. Here, gear 1 is driver and gear 2 is driven gear.
2. Draw rotation direction of each gear. Here, gear 1 rotates anticlockwise, therefore gear 2 rotates clockwise.
3. Draw tangential force on **driven gear** first. Here, tangential force on gear 2 is towards up. (Follow direction of rotation)
4. Draw radial force, F_r which is opposite to each other.
5. Draw tangential force on **driver gear** which is opposite to driven gear. Here, tangential force on gear 1 is towards down.



Find torque on gear 1

$$P = \frac{2\pi nT}{60}$$

$$80000 = \frac{2\pi \times 1600 \times T_1}{60}$$

$$T_1 = 477.46 Nm$$

Radius of gear 1, $r_1 = \frac{d_1}{2} = 0.054m$

From torque equation, we have

$$T_1 = F_{t1} \times r_1$$

$$F_{t1} = \frac{T_1}{r_1} = \frac{477.46}{0.054} = 8841.85N$$

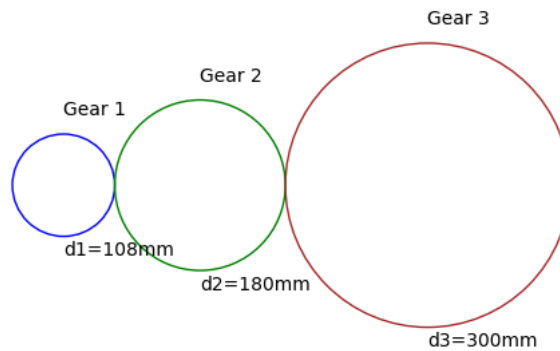
$$F_{t1} = F_{t2} = 8841.85N$$

Radial force on gear 1,

$$\begin{aligned} F_{r1} &= F_{t1} \tan \phi \\ F_{r1} &= 8841.85 \times \tan 20^\circ \\ &= 3218.17N \end{aligned}$$

$$F_{r1} = F_{r2} = 3218.17N$$

Gear 2 and 3



$$\text{Radius of gear 2, } r_2 = \frac{d_2}{2} = 0.09m$$

Find speed of gear 2

$$\begin{aligned} \frac{N_1}{N_2} &= \frac{n_2}{n_1} \\ \frac{18}{30} &= \frac{n_2}{1600} \\ n_2 &= 960rpm \end{aligned}$$

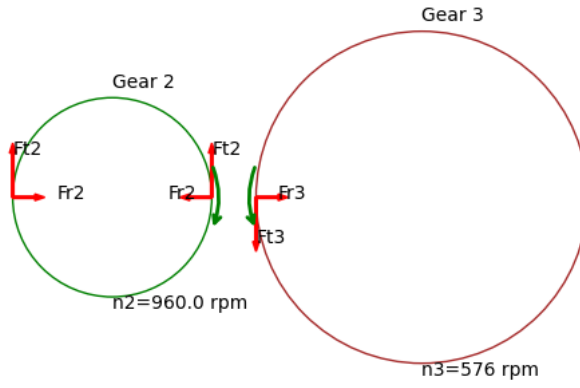
Draw FBD between gear 2 and gear 3

Find torque on gear 2

$$\begin{aligned} P &= \frac{2\pi nT}{60} \\ 80000 &= \frac{2\pi \times 960 \times T_2}{60} \\ T_2 &= 795.77Nm \end{aligned}$$

$$\text{Radius of gear 2, } r_2 = \frac{d_2}{2} = 0.09m$$

FBD between Gear 2 and 3



From torque equation, we have

$$\begin{aligned}
 T_2 &= F_{t2} \times r_2 \\
 F_{t2} &= \frac{T_2}{r_2} \\
 &= \frac{795.77}{0.09} \\
 &= 8841.89N
 \end{aligned}$$

$$F_{t2} = F_{t3} = 8841.89N$$

Radial force on gear 2,

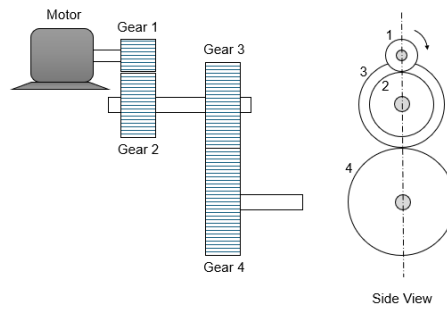
$$\begin{aligned}
 F_{r2} &= F_{t2} \tan \phi \\
 F_{r2} &= 8841.89 \times \tan 20^\circ \\
 &= 3218.18N
 \end{aligned}$$

$$F_{r2} = F_{r3} = 3218.18N$$

Question 2

In gear train shown in Figure Q2, gear 1 is the inlet of the gear train and transfers 1.8kW of power at 2400 rpm. The module of all gears in the train drive is 4 mm and the pressure angle is 20° . Diameter for gear 1, 2, 3 and 4 are 32mm, 60mm, 80mm and 100mm respectively.

- (i) Determine the number of teeth for gears 1, 2, 3 and 4.
- (ii) Determine the tangential and radial forces between gears 1 and 2 and between gears 3 and 4.
- (iii) If the diameter of gear 4 is increased to 120mm, explain how it affects the torque of gear 4



Question 3

The gear train in Figure Q3 is required to transmit the power of 40 kW at 1500 rpm from pinion 1. The ratio of speed between gear 1 and 2 is 5:2. Due to limited space, the ratio number of teeth on gear 3 to gear 4 is 1:3 with module of 4 mm and a pressure angle of 20°.

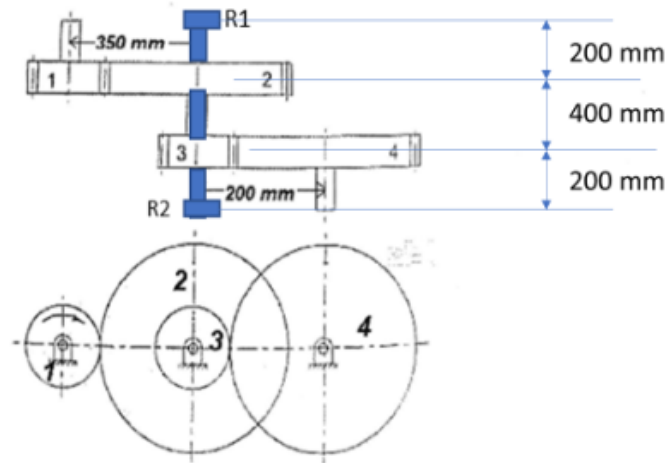


Figure Q3

- (i) Determine the number of teeth on gears 1, 2, 3 and 4
- (ii) Determine the radii of gears 1, 2, 3 and 4
- (iii) Calculate the speed on gears 2, 3 and 4
- (iv) Determine the tangential and radial forces on each gear.
- (v) Calculate the magnitude of reaction at the shaft of gear 2 and 3.
- (vi) Calculate the reaction at bearing R1 and R2.

Well-defined Questions

- (i) If the gear number 4 is reduced to 65, how does it affect gear 4 torque and speed?
- (ii) What happens to the gear 4 speed and torque if the teeth on gear 2 are doubled?

Answer**Q2**

- (i) $N_1 = 8, N_2 = 15, N_3 = 20, N_4 = 25$
- (ii) $F_{t1} = F_{t2} = 447.5N, F_{r1} = F_{r2} = 162.88N$
 $F_{t3} = F_{t4} = 335.75N, F_{r3} = F_{r4} = 122.2N$
- (iii) When the diameter of gear 4 is increased, the torque of gear 4 also increases from 16.79 Nm to 20.14 Nm.

Q3

- 1. $N_1 = 50, N_2 = 125, N_3 = 25, N_4 = 75$
- 2. $r_1 = 100mm, r_2 = 250mm, r_3 = 50mm, r_4 = 150mm$
- 3. $n_2 = 600rpm, n_3 = 600rpm, n_4 = 200rpm$
- 4. $F_{t1} = F_{t2} = 2546.5N, F_{r1} = F_{r2} = 926.85N$
 $F_{t3} = F_{t4} = 12732.4N, F_{r3} = F_{r4} = 4634.21N$
- 5. $15722.4N, \theta = 76.3^\circ$
- 6. $R1=5114.02N, R2=10690N$

Well-defined Questions

- (i) When the number of teeth of gear 4 is reduced, the speed of gear 4 increases from 200 rpm to 231 rpm, while the torque of gear 4 decreases from 1909.86 Nm to 1655.21 Nm
- (ii) When the number of teeth on gear 2 is doubled, the speed of gear 4 decreases from 200 rpm to 100 rpm, while the torque of gear 4 increases from 1909.86 Nm to 3819.72 Nm.